Cavity blockade: a way to make photons interact

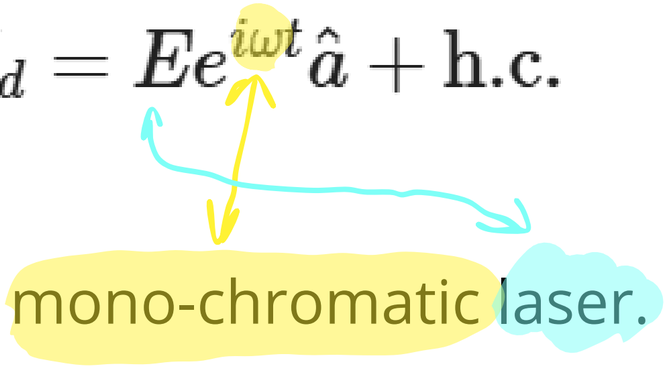
Leonardo Fardin

The system

A Fabry-Perrot cavity with an atom inside.
The system is derscribed by the Jaynes-Cumming Hamiltonian:

$$\hat{H}(t) = \hbar\omega_c \hat{a}^\dagger \hat{a} + \hbar\omega_a |e\rangle\langle e| + \hbar\tilde{g}(\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger) + \hat{H}_d$$

$$\hat{H}_d = E e^{i\omega t} \hat{a} + \text{h.c.}$$



The drive can be realized with a mono-chromatic laser.

We can move to a rotating frame according to the laser's frequency and we obtain:

$$R(t) = \exp\left[i\omega t(|e\rangle\langle e| + \hat{a}^\dagger \hat{a})\right]$$

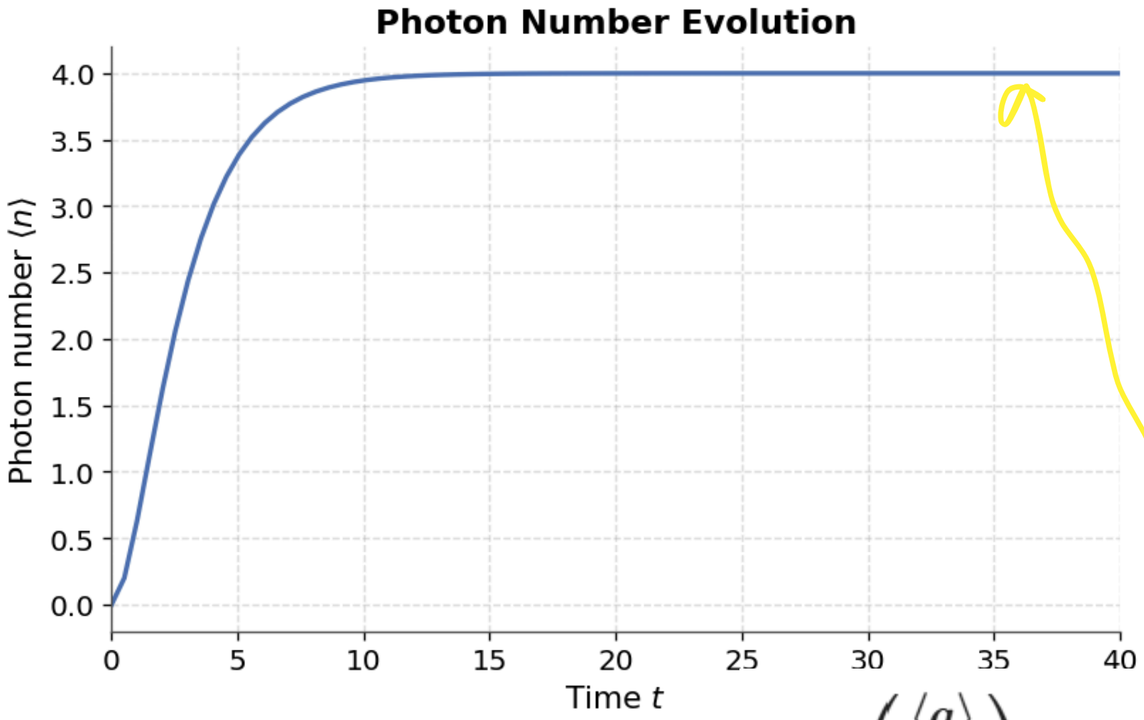
$$\hat{H}'(t) = \hat{H}' = \hbar(\omega_c - \omega)\hat{a}^\dagger \hat{a} + \hbar(\omega_a - \omega)|e\rangle\langle e| + \hbar\tilde{g}(\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger) + \varepsilon(\hat{a} + \hat{a}^\dagger)$$

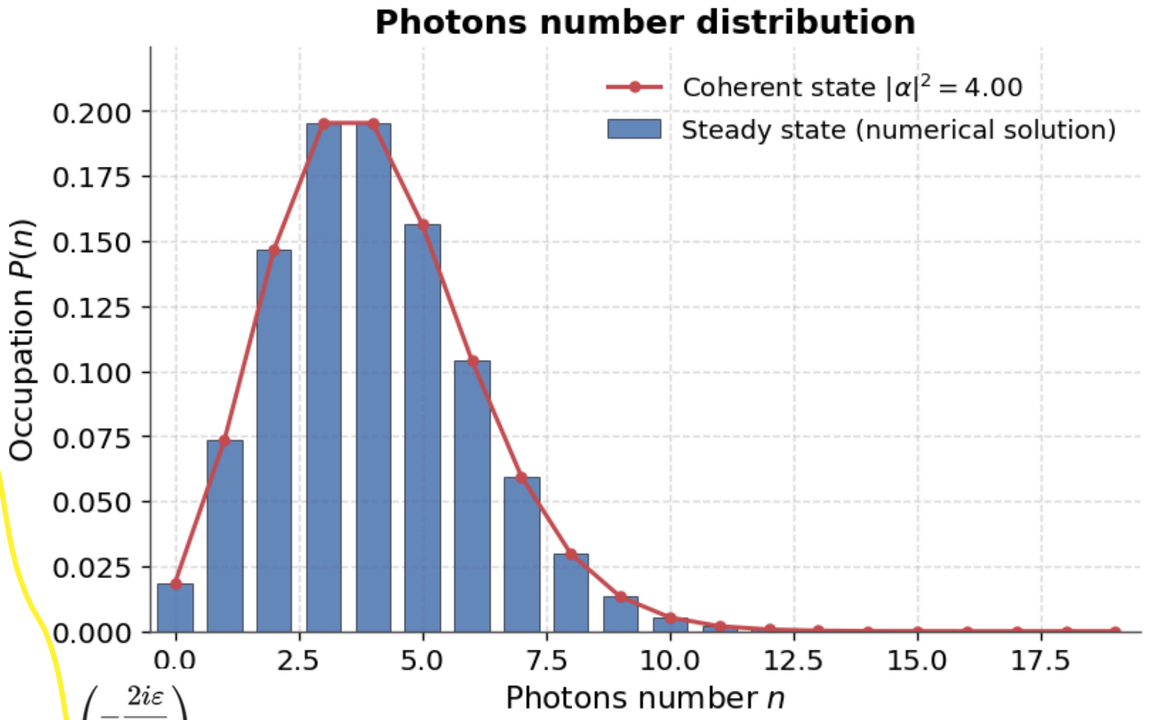
Uncoupled Hamiltonian

We tune the laser with the cavity and remove the atom-cavity coupling:

For $\omega = \omega_a = \omega_c \equiv \omega_0$ and $\tilde{g} = 0$:

$$\hat{H}' = \varepsilon(\hat{a} + \hat{a}^\dagger)$$





$$\vec{S} = \begin{pmatrix} \langle a \rangle \\ \langle a^\dagger \rangle \\ \langle N \rangle \end{pmatrix} \qquad \vec{S}_{steady} = \begin{pmatrix} -\frac{2i\varepsilon}{\hbar\Gamma} \\ \frac{2i\varepsilon}{\hbar\Gamma} \\ \frac{4\varepsilon^2}{\hbar^2\Gamma^2} \end{pmatrix}$$

Uncoupled Hamiltonian

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```
N = 20                          #hilbert space dim
a = destroy(N)                  #ladder operator for the light

kappa = 1.0          # we fix the target average number and the decay rate k
n_target = 4.0       # and using the results of the bloch optical equation,
epsilon = kappa * np.sqrt(n_target) / 2   # we infer the value of epsilon

H = epsilon * (a + a.dag())                 #the hamiltonian
c_ops = [np.sqrt(kappa) * a]                #collapse operators of the cavity

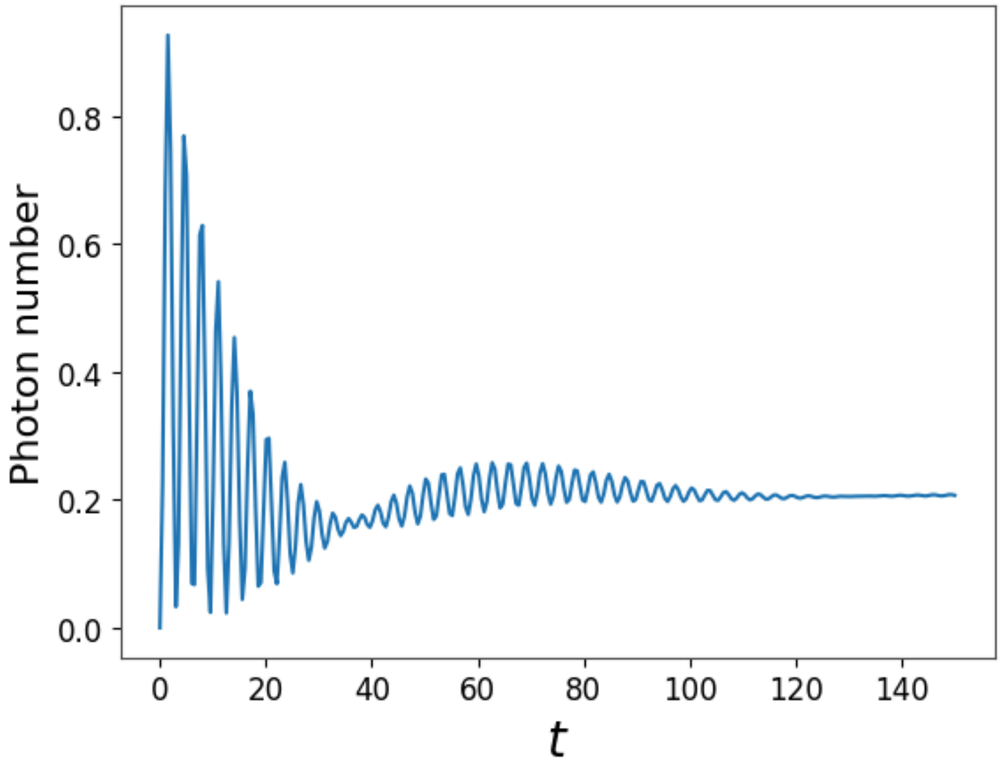
psi0 = fock(N, 0)                           #initial state

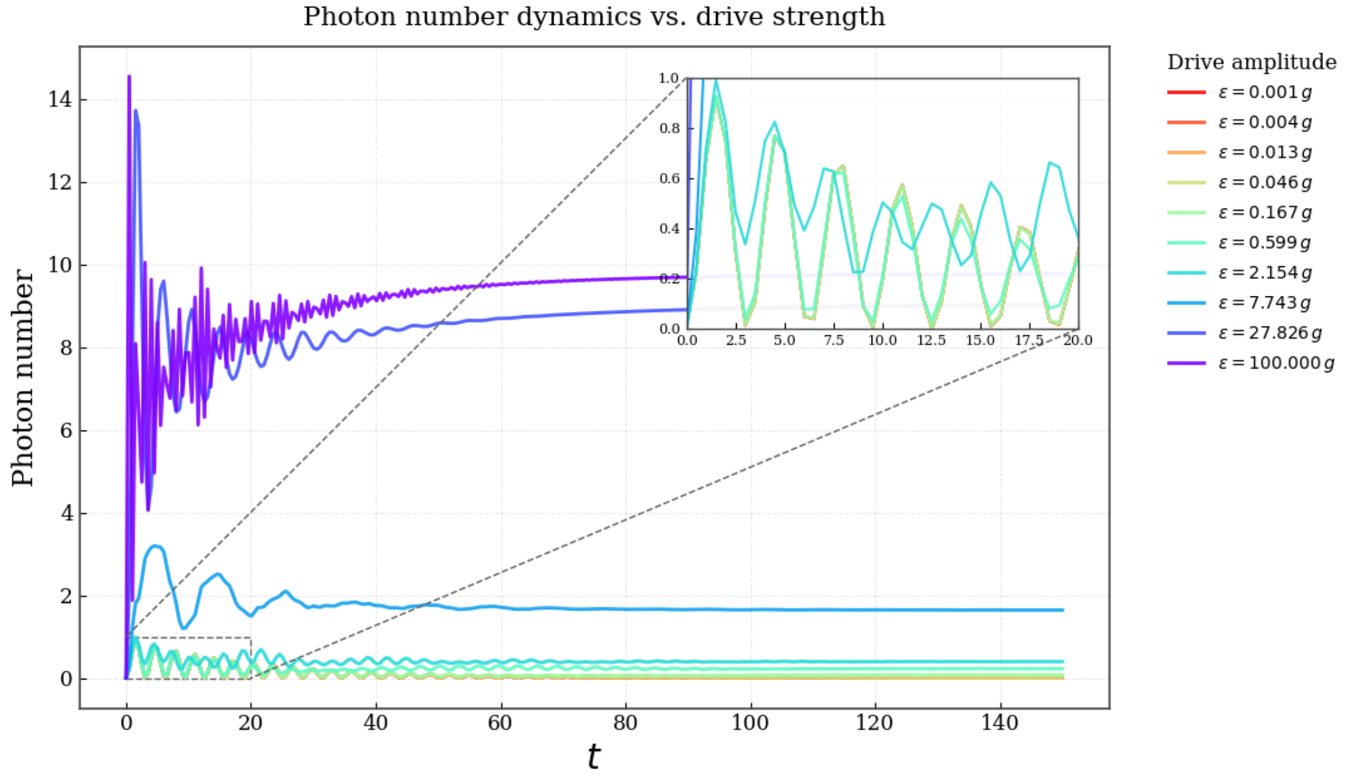
tlist = np.linspace(0, 40, 80)
result = mesolve(H, psi0, tlist, c_ops, e_ops=[a.dag() * a],
                 options={"store_states": True})
```

Laser resonant with Lower dressed state

We tune the laser with the cavity and remove the atom-cavity coupling:

$$\omega = \omega_0 - g \qquad \hat{H}' = \hbar\tilde{g}\hat{a}^\dagger\hat{a} + \hbar\tilde{g}|e\rangle\langle e| + \hbar\tilde{g}(\hat{\sigma}_+\hat{a} + \hat{\sigma}_-\hat{a}^\dagger) + \varepsilon(\hat{a} + \hat{a}^\dagger)$$





Laser resonant with Lower dressed state treating the drive in a perturbative way

$$H = \hbar g(a^\dagger a + |e\rangle\langle e| + \sigma_+ a + \sigma_- a^\dagger) + \varepsilon(a + a^\dagger) = H_0 + \varepsilon V$$

$$\Longrightarrow\ H^{(1)} = 0\ \Longrightarrow\ \text{No first order correction}$$

$|g,0\rangle\ (E=0), \qquad |+\rangle^{(2)} = \frac{1}{\sqrt{2}}(|g,2\rangle + |e,1\rangle)\ \left(E_+^{(2)} = \sqrt{2}\,\hbar g(\sqrt{2}+\tfrac{1}{2})\right)$

$|-\rangle^{(2)} = \frac{1}{\sqrt{2}}(-|g,2\rangle + |e,1\rangle)\ \left(E_-^{(2)} = \sqrt{2}\,\hbar g(\sqrt{2}-\tfrac{1}{2})\right)$

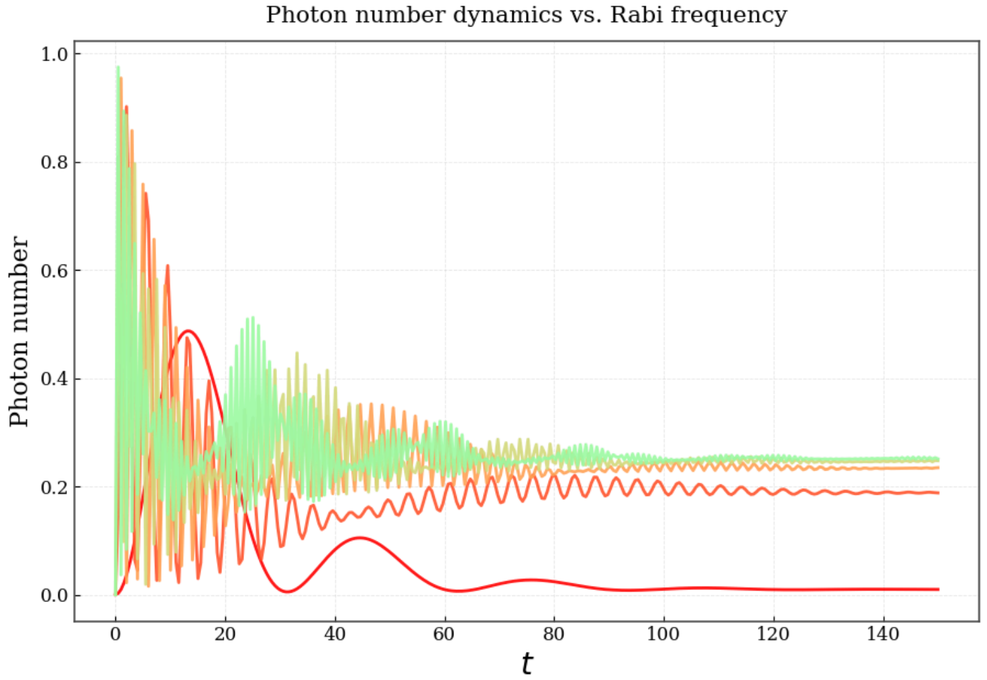
$$H_{+-}^{(2)} \approx\approx -1.33\,\frac{\varepsilon^2}{g}$$

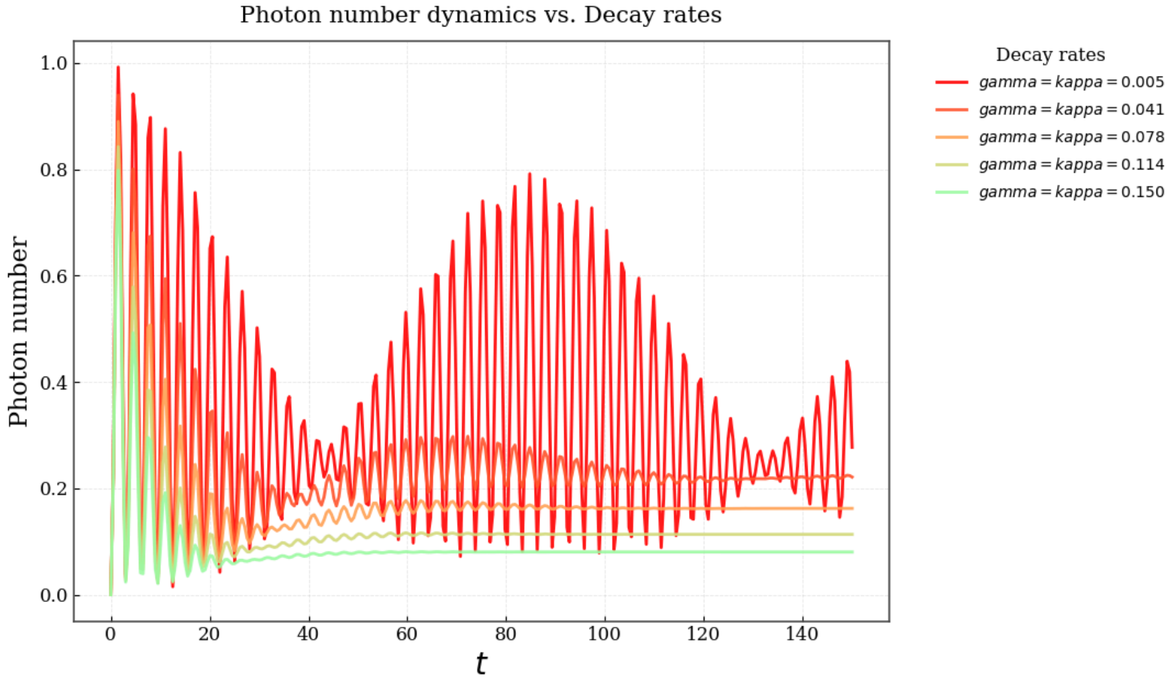
Let's define α so that $\varepsilon = \alpha g$ where $\alpha \ll 1$.

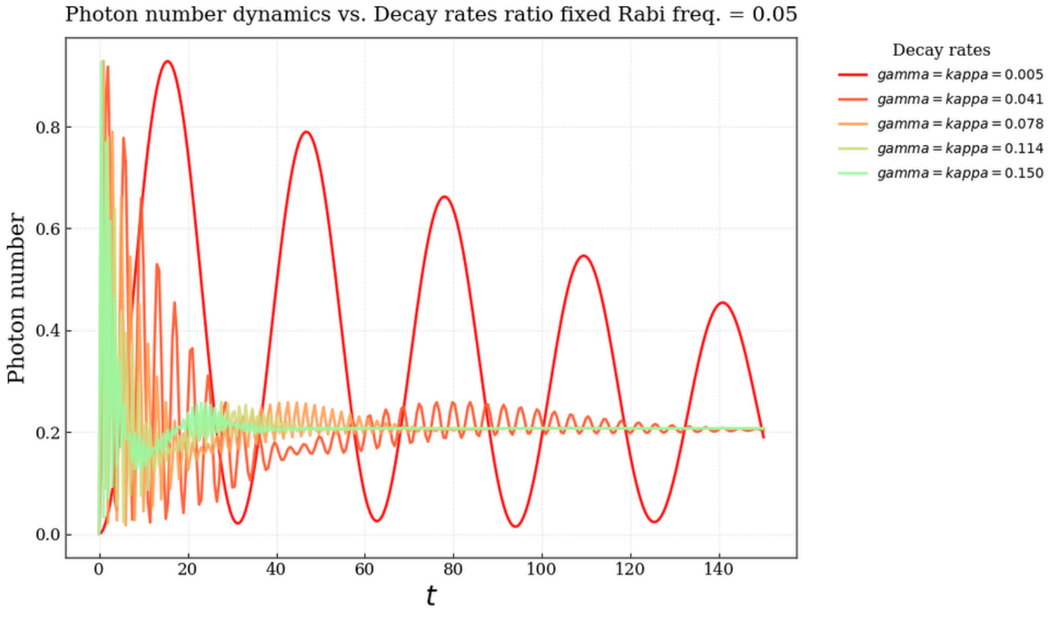
$$H_{+-}^{(2)} \sim \frac{\varepsilon^2}{g} = \alpha^2 g, \qquad \Delta E = E_+^{(1)} - E_-^{(1)} = \hbar g$$

$$\frac{H_{+-}^{(2)}}{\Delta E} \sim \alpha^2 \ll 1$$

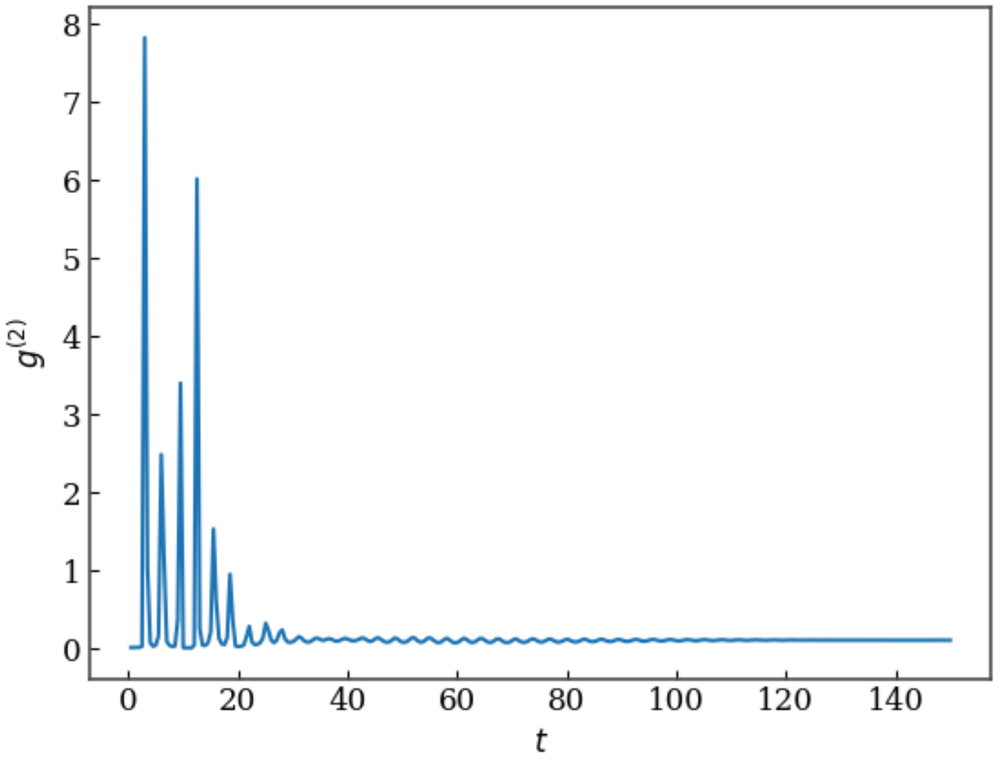
Laser resonant with Lower dressed state

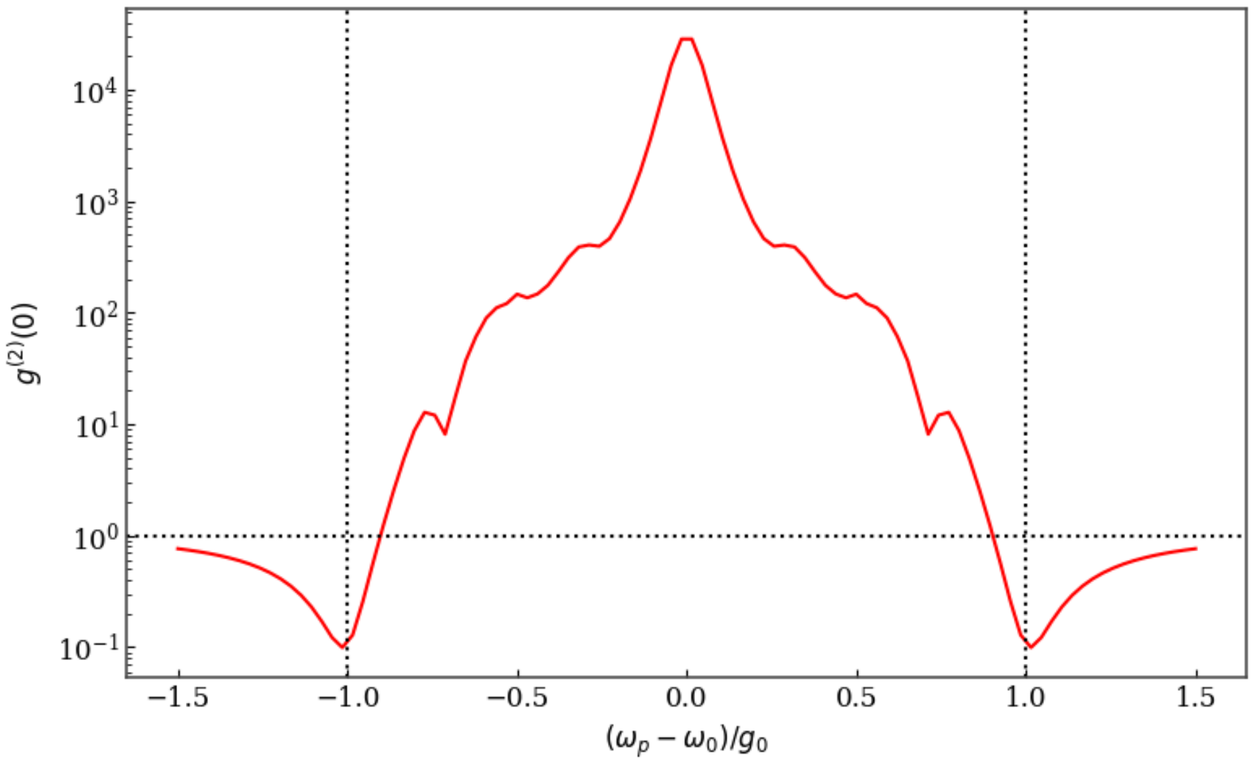






Laser resonant with Lower dressed state





Analytical derivation for the steady states

Uncoupled

$$\dot{\rho} = \frac{i}{\hbar}[\rho, H] + \frac{\Gamma}{2}\left[2a\rho a^\dagger - \{a^\dagger a, \rho\}\right]$$

$$\frac{d}{dt}\langle a^\dagger a\rangle = \frac{d}{dt}\langle N\rangle = \frac{i\varepsilon}{\hbar}\left(\langle a\rangle - \langle a^\dagger\rangle\right) - \Gamma\langle N\rangle$$

$$\frac{d}{dt}\langle a\rangle = -\frac{i\varepsilon}{\hbar} - \frac{\Gamma}{2}\langle a\rangle$$

$$\frac{d}{dt}\langle a^\dagger\rangle = \frac{i\varepsilon}{\hbar} - \frac{\Gamma}{2}\langle a^\dagger\rangle$$

$$\vec{S}_{steady} = \begin{pmatrix} -\dfrac{2i\varepsilon}{\hbar\Gamma} \\ \dfrac{2i\varepsilon}{\hbar\Gamma} \\ \dfrac{4\varepsilon^2}{\hbar^2\Gamma^2} \end{pmatrix}$$

Low dressed state resonant

$$\begin{pmatrix} \langle\dot{a}\rangle \\ \langle\dot{a}^\dagger\rangle \\ \langle\dot{\sigma}_-\rangle \\ \langle\dot{\sigma}_+\rangle \\ \langle\dot{\sigma}_z\rangle \\ \langle\dot{a^\dagger a}\rangle \\ \langle\dot{N}\rangle \end{pmatrix} = \begin{pmatrix} -\frac{\kappa}{2} & 0 & -ig & 0 & 0 & 0 & 0 \\ 0 & -\frac{\kappa}{2} & 0 & ig & 0 & 0 & 0 \\ 0 & 0 & ig - \frac{\Gamma}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -ig - \frac{\Gamma}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\Gamma & 0 & 0 \\ \frac{i\varepsilon}{\hbar} & -\frac{i\varepsilon}{\hbar} & 0 & 0 & 0 & -\kappa & 0 \\ \frac{i\varepsilon}{\hbar} & -\frac{i\varepsilon}{\hbar} & 0 & 0 & -\frac{\Gamma}{2} & -\kappa & 0 \end{pmatrix} \begin{pmatrix} \langle a\rangle \\ \langle a^\dagger\rangle \\ \langle\sigma_-\rangle \\ \langle\sigma_+\rangle \\ \langle\sigma_z\rangle \\ \langle a^\dagger a\rangle \\ \langle N\rangle \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ ig\langle a\rangle\langle\sigma_z\rangle \\ -ig\langle a^\dagger\rangle\langle\sigma_z\rangle \\ 2ig\left(\langle\sigma_-\rangle\langle a^\dagger\rangle - \langle\sigma_+\rangle\langle a\rangle\right) \\ ig\left(\langle\sigma_+\rangle\langle a\rangle - \langle\sigma_-\rangle\langle a^\dagger\rangle\right) \\ 0 \end{pmatrix} + \begin{pmatrix} -\frac{i\varepsilon}{\hbar} \\ \frac{i\varepsilon}{\hbar} \\ 0 \\ 0 \\ -\Gamma \\ 0 \\ -\frac{\Gamma}{2} \end{pmatrix}$$

Approximation for closed form

$$\langle\sigma_+ a\rangle = \langle\sigma_+\rangle\langle a\rangle \text{ and } \langle\sigma_- a^\dagger\rangle = \langle\sigma_-\rangle\langle a^\dagger\rangle$$

Non-linear approximated eqation that has to be solved with numerical methods